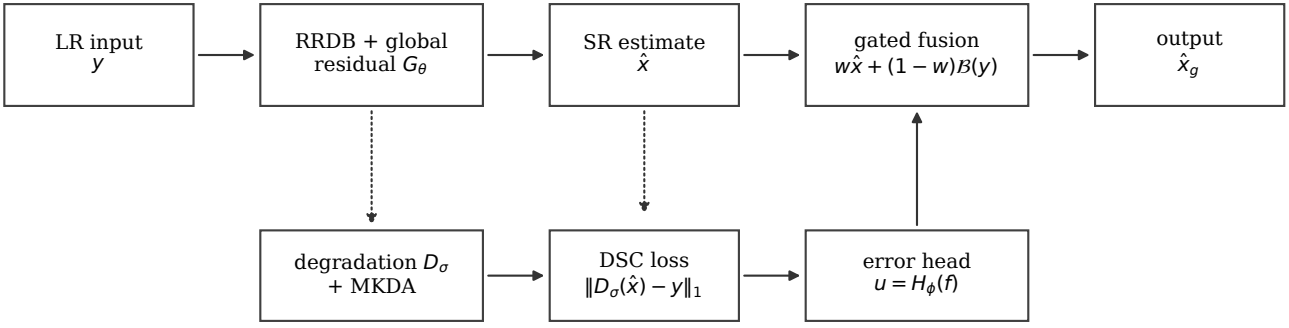(a) training and inference